

Notes on Three Classic Strategy Games: Colonel Blotto, Public Goods, and Iterated Prisoner's Dilemma

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Abstract

I cover the game theory behind three multiplayer strategy games: Colonel Blotto (resource allocation with simultaneous competition), the Public Goods Game (social dilemmas and free-riding), and Axelrod's Iterated Prisoner's Dilemma Tournament (the evolution of cooperation). Each game illustrates a different fundamental tension in strategic interaction. These are my working notes—I try to give the intuition first and the formalism second.

1 Colonel Blotto

1.1 The Setup

Colonel Blotto is one of those games that's trivially easy to describe and absurdly hard to solve. Two players each have N troops to allocate across K battlefields. Allocations are simultaneous and secret. Whoever sends more troops to a battlefield wins it, and the player who wins a majority of battlefields wins the game.

Formally, player 1 chooses $(x_1, x_2, \dots, x_K)$ with $\sum_k x_k = N$ and $x_k \geq 0$, and player 2 chooses $(y_1, \dots, y_K)$ similarly. Player 1 wins battlefield k if $x_k > y_k$. The player winning more than $K/2$ battlefields wins.

1.2 No Pure Strategy Nash Equilibrium

Here's the key fact: **there is no pure strategy Nash equilibrium in Colonel Blotto**. Every deterministic allocation can be beaten. If you put all your troops in the first $\lceil K/2 \rceil$ battlefields, I put one troop more than you in those same battlefields and zero elsewhere. If you spread evenly, I concentrate on just over half the battlefields.

I spent a while trying to find a “good” fixed allocation before accepting this. The game forces you to randomize.

1.3 Nash Equilibrium in the Continuous Case

In the symmetric continuous version (both players have the same budget, troops are divisible), the unique Nash equilibrium involves randomizing uniformly over all allocations that satisfy the budget constraint. Specifically, the NE marginal distribution on each battlefield is uniform on $[0, 2N/K]$.

One clean way to think about this: sample a random allocation from the **symmetric Dirichlet distribution** $\text{Dir}(1, 1, \dots, 1)$ over K categories and multiply by N . This produces a uniform random point on the simplex $\{x : \sum x_k = N, x_k \geq 0\}$.

Why does uniform randomization work? Notice that against a uniform opponent, every allocation wins exactly half the time (by symmetry). So no deviation can improve your win rate—you're

already at 50% against any uniform opponent, and they're at 50% against you. That's an equilibrium.

1.4 The Discrete Case

The discrete version (integer troops) is much harder. For small N and K , you can enumerate all allocations and solve the resulting zero-sum matrix game via linear programming. For larger instances, the number of allocations grows as $\binom{N+K-1}{K-1}$, which gets out of hand quickly. Approximation methods and the Dirichlet sampling approach become necessary.

2 Public Goods Game

2.1 The Social Dilemma

The Public Goods Game is the n -player generalization of the Prisoner's Dilemma, and it captures one of the most important problems in economics and political science: why do people free-ride on public goods?

Each of N players receives an endowment E and secretly chooses a contribution $c_i \in [0, E]$ to a common pool. The pool is multiplied by a factor M and divided equally among all N players. Player i 's payoff is:

$$\pi_i = (E - c_i) + \frac{M \cdot \sum_{j=1}^N c_j}{N}$$

The first term is what you kept. The second is your share of the public good.

2.2 The MPCR and the Dilemma

The **Marginal Per-Capita Return** (MPCR) is the key parameter:

$$\text{MPCR} = \frac{M}{N}$$

This tells you how much you get back per dollar you contribute. If I contribute \$1 more, the pool grows by \$1, gets multiplied by M , and I get back M/N of that. My net return on that dollar is $M/N - 1$.

The social dilemma arises when $1 < M < N$, i.e., $\text{MPCR} < 1$:

- **Individually**, contributing is irrational: you put in \$1 and get back $M/N < \$1$. You'd rather keep your money.
- **Collectively**, contributing is efficient: each dollar contributed generates M dollars of total value, which exceeds the \$1 cost since $M > 1$.

2.3 Nash Equilibrium

The Nash equilibrium is easy to compute and depressing:

$$c_i^* = 0 \quad \text{for all } i$$

Nobody contributes. To see this, notice that player i 's payoff is linear in c_i with slope $-1 + M/N < 0$ (when $M < N$). So regardless of what others do, player i maximizes their payoff by setting $c_i = 0$.

But the socially optimal outcome is $c_i = E$ for all i , giving each player a payoff of ME versus E at the Nash equilibrium. Since $M > 1$, everyone would be better off if everyone contributed. This is the textbook case of a **Pareto-dominated Nash equilibrium**.

2.4 The Gap

The “tragedy” can be quantified. At Nash ($c_i = 0$), each player gets E . At the social optimum ($c_i = E$), each player gets ME . The efficiency loss is:

$$\text{Efficiency ratio} = \frac{E}{ME} = \frac{1}{M}$$

With $M = 3$ and $N = 5$, for example, the Nash outcome captures only 1/3 of the socially optimal surplus. That’s a lot of value left on the table.

3 Iterated Prisoner’s Dilemma Tournament

3.1 The Stage Game

The Prisoner’s Dilemma has a 2×2 payoff matrix where each player simultaneously chooses Cooperate (C) or Defect (D):

	C	D
C	R, R	S, T
D	T, S	P, P

with the constraints $T > R > P > S$ and $2R > T + S$. The standard values are $T = 5$, $R = 3$, $P = 1$, $S = 0$.

The one-shot game has a unique Nash equilibrium: both defect (getting P each), even though mutual cooperation (R each) would be better. Defection is a dominant strategy—no matter what your opponent does, you’re better off defecting.

3.2 Axelrod’s Tournament

Robert Axelrod’s (1984) insight was to ask: what happens when the same players meet repeatedly? In an iterated PD with sufficiently many rounds, cooperation can be sustained because defection can be punished in future rounds.

He organized a tournament where researchers submitted strategies for the iterated PD. Each strategy played every other strategy in a round-robin format over many rounds. The total score determined the winner.

3.3 Tit-for-Tat

The winning strategy—submitted by Anatol Rapoport—was embarrassingly simple: **Tit-for-Tat** (TFT). Cooperate on the first move, then copy whatever the opponent did last round.

TFT has four properties that Axelrod identified as critical:

1. **Nice**: never defects first. This avoids provoking unnecessary conflict.
2. **Retaliatory**: immediately punishes defection. This deters exploitation.
3. **Forgiving**: goes back to cooperating as soon as the opponent cooperates. This prevents death spirals.
4. **Clear**: its behavior is easy for opponents to understand and predict, facilitating cooperation.

The deep result from Axelrod’s tournament is that **“nice” strategies (those that never defect first) systematically outperformed “mean” strategies**. This seems counterintuitive—nice guys finish last, right? But in a round-robin, mean strategies beat each other up and drag each other’s scores down, while nice strategies cooperate with each other and achieve high mutual scores.

3.4 Pavlov: Win-Stay, Lose-Shift

After playing with this for a while, I noticed that TFT has a weakness: it can’t exploit unconditional cooperators (Always Cooperate). TFT just cooperates back, getting R every round, but it could potentially do better.

Pavlov (also called Win-Stay, Lose-Shift) addresses this. The rule is:

- If the last round’s outcome was “good” (CC or DC, meaning you got R or T), repeat your last action.
- If the last round’s outcome was “bad” (CD or DD, meaning you got S or P), switch your action.

Pavlov cooperates with cooperators (like TFT), retaliates against defectors (like TFT), but also *exploits* unconditional cooperators by occasionally defecting and then cooperating again when punished. In some tournament formats, Pavlov outperforms TFT.

3.5 Formal Analysis: Discount Factor

In the infinitely repeated PD with discount factor δ (probability of continuing to the next round), TFT sustains mutual cooperation as a Nash equilibrium if and only if:

$$\frac{R}{1-\delta} \geq T + \frac{P \cdot \delta}{1-\delta}$$

which simplifies to:

$$\delta \geq \frac{T-R}{T-P}$$

With the standard payoffs ($T = 5, R = 3, P = 1$), this gives $\delta \geq 1/2$. So as long as there’s at least a 50% chance of meeting again, TFT can sustain cooperation. **The shadow of the future is long enough to make cooperation rational.**

4 Connections

All three games share a common theme: **the tension between individual rationality and collective welfare**. In Colonel Blotto, every pure strategy is exploitable. In the Public Goods Game, the individually rational action (free-ride) is collectively disastrous. In the Prisoner’s Dilemma, defection dominates but mutual cooperation is better for everyone.

The resolution in each case is different: Blotto requires randomization, Public Goods requires mechanisms or norms to enforce contribution, and the iterated PD shows that reputation and repeated interaction can sustain cooperation naturally. It’s easy to see that these map onto real-world problems in security (Blotto), public policy (Public Goods), and diplomacy (PD).

References

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